

ABU HASNAT ABDULLAH

Senior AI Engineer · LLM · RAG · MLOps

Dhaka, Bangladesh · +8801710608387 · abdullah.2010bd@gmail.com · linkedin.com/in/hasnatabdullah · github.com/hasnat-abdullah

PROFESSIONAL SUMMARY

Senior Software Engineer with **10+ years of professional experience**, including **6+ years** architecting production-grade AI systems for enterprise clients across the **Nordics, Sweden, and Bangladesh**. Ship LLM applications end-to-end — RAG pipelines, fine-tuned domain models (LoRA / QLoRA), and MLOps platforms on AWS and Azure — with measurable impact: lifted industrial CV detection accuracy **from 40% to 98%**, fine-tuned a local 3B LLM to **97% accuracy**, and led an **11-engineer team** across 6 concurrent enterprise projects. Obsessed with responsible-AI guardrails, evaluation discipline, and turning research-grade ideas into reliable production systems.

TECHNICAL SKILLS

Generative AI & LLMs OpenAI GPT, Anthropic Claude, Llama 3 / 3.2, Mistral, DeepSeek, Qwen; prompt engineering, RAG architectures, semantic chunking, evaluation frameworks, hallucination guardrails

Fine-Tuning LoRA, QLoRA, PEFT, Unsloth, HuggingFace Transformers, domain adaptation

Vector & Knowledge PGVector, Chroma, Neo4j knowledge graphs, embedding optimization

ML & Computer Vision PyTorch, TensorFlow, scikit-learn, YOLOv11, OpenCV, OCR, Pandas, NumPy

MLOps & Data Engineering Mage.ai (ETL / orchestration), MLflow, AzureML, model registry, feature stores, drift monitoring, CI/CD

Languages Python (primary), SQL, JavaScript, HTML, CSS

Backend & APIs FastAPI, Django, Flask, REST APIs, async services, microservices, Kafka, Redis

Databases PostgreSQL, MongoDB, InfluxDB (time-series)

Cloud & DevOps AWS (Lambda, EC2, ECS, S3, RDS, API Gateway), Azure, Docker

Leadership & Delivery Cross-functional team leadership (11+ engineers), system design, technical mentorship, agile delivery, stakeholder management

PROFESSIONAL EXPERIENCE

Senior Software Engineer · Cefalo Bangladesh Ltd. · Sensa AS, Norway *Nov 2024 — Present*

- Own AI engineering for Sensa, a Norwegian industrial-AI company — delivering LLM applications, computer-vision systems, time-series MLOps, and the underlying data platform end-to-end, with deep focus on evaluation, observability, and responsible-AI guardrails.
- Unity AI** — Architected an LLM-powered conversational assistant that lets industrial operators query plant data in natural language. Designed a custom RAG pipeline combining a **Neo4j knowledge graph** with an **InfluxDB time-series store**, with grounded citations and data-governance rules that drive confident, source-backed answers and minimize hallucinations.

› LLM, OpenAI, RAG, FastAPI, Neo4j, InfluxDB, PGVector, Pandas, SaaS, Azure

- **Unity Engine** — Built Sensa's internal **data-orchestration and feature platform** — the foundation powering Unity AI, Unity-MLOps, and Pellet Detection. Designed **Mage.ai ETL pipelines** to ingest sensor and external data into time-series and feature stores, with reusable transforms that serve both real-time anomaly detection and downstream RAG retrieval.
› *Mage.ai, Python, FastAPI, InfluxDB, Neo4j, PGVector, Pandas, Azure*
- **CV Analyser** — Shipped an AI résumé-parsing platform that extracts text from CVs and returns structured, ATS-ready output with role/skill categorization, plus a chatbot for recruiter Q&A over the parsed corpus. Fine-tuned **Llama 3.2, DeepSeek, and Qwen** with Unsloth (QLoRA), benchmarked trade-offs across model families, and reached **97% accuracy** on a local 3B model — matching cloud-API quality at a fraction of the cost.
› *Llama 3.2, DeepSeek, Qwen, Unsloth, QLoRA / PEFT, HuggingFace, FastAPI, OCR, Python*
- **Fish Food Pellet Detection** — Productionized a **YOLOv11-nano** computer-vision model that detects and counts fish-food pellets in real-time aquaculture footage, lifting detection accuracy **from 40% to 98%** and unlocking automated feed-rate optimization for sensor-equipped fish farms.
› *YOLOv11, PyTorch, OpenCV, Python*
- **Unity-MLOps** — Engineered an end-to-end anomaly-detection pipeline on time-series sensor data with automated training, model registry, deployment, and drift monitoring — turning manual log review into proactive, real-time alerts.
› *FastAPI, Python, scikit-learn, AzureML, MLflow*

Senior Software Engineer · Strativ AB · Apr 2021 — Jan 2024

- Led a cross-functional team of **11 engineers** across **6 concurrent enterprise projects** for top-tier Swedish clients — owning system design, backend architecture (Python, Django), code review, and delivery, while staying hands-on with API and AI feature development.
- **TourGPT** — Architected a SaaS RAG chatbot grounded in a custom tour knowledge base, delivering personalized recommendations and end-to-end order management. Owned prompt engineering, embedding strategy, and safety guardrails **from prototype to production**.
› *LLM, OpenAI, RAG, Django, PostgreSQL, PGVector, SaaS, AWS*
- **Investment Management System** — Designed and shipped a portfolio-analytics platform tracking performance, sales, and accounting across multiple portfolio companies, integrating third-party data sources via event-driven processing.
› *Django, REST API, PostgreSQL, SaaS, Kafka, AWS*
- **Health Check** — Delivered a digital primary-healthcare platform ingesting physical measurements and lab blood-test reports via third-party APIs, surfacing actionable health metrics for end users.
› *Django, REST API, PostgreSQL, Kafka, NumPy, AWS*

Software Engineer · Circle Fintech Ltd. · Sep 2020 — Apr 2021

- Shipped Boarding Bay, an eKYC platform for **Jamuna Bank**, modernizing customer onboarding and accelerating business financial transaction processing across branches.
- **Boarding Bay** — Built 3 production-grade microservices (NID OCR, face detection, billing) that automated identity verification and significantly reduced manual onboarding time.
› *FastAPI, Flask, Django, REST API, Microservices, TensorFlow, PyTorch, Pandas, NumPy*

Junior Research Analyst · Economic Research Group *Apr 2020 — Aug 2020*

- Analyzed mobile-app data for Romoni (on-demand beauty marketplace) to drive sales growth and forecast demand using regression models, informing pricing and marketing decisions.

▸ *Python, PyTorch, NumPy, Pandas, Matplotlib, Regression*

Founder & CEO · Shikhte Chai (Educational Initiative) · *Apr 2016 — Jan 2020*

- Founded and scaled an educational initiative in Bangladesh, leading strategy, operations, and program design to upskill students in technology and entrepreneurship.

EDUCATION

B.Sc. in Computer Science & Engineering · North South University, Bangladesh

2015 — 2019

- CGPA: 3.07 / 4.00
- Relevant Coursework: Artificial Intelligence, Pattern Recognition & Neural Networks, Data Structures, Design and Analysis of Algorithms, Probability and Statistics, Microeconomics.

CERTIFICATIONS

Neo4j GraphAcademy *2025*

- **Neo4j Fundamentals** · **Cypher Fundamentals** · **Neo4j & Generative AI Fundamentals** — graph databases, Cypher query language, and grounded RAG with knowledge graphs.

Data Science · **Prof. Jennifer Widom, Stanford University** *2024*

- Machine learning (regression, classification, clustering), data manipulation, mining, and visualization with Tableau, Python, and R.

Business Analysis & Prototyping · **LICT (Bangladesh Govt.) & BracIT** *2019*

- 3-month program in requirements analysis, strategic planning, design thinking, process improvement, and stakeholder communication.

RESEARCH PUBLICATIONS

- **Predicting the Result of a Cricket Match by Applying Data Mining Techniques** (2020) — Applied Recursive Feature Elimination, ZeroR, Decision Tree, Random Forest, and XGBoost to predict match outcomes. Published on ResearchGate.